

Clark Zhang

 Email  LinkedIn  Portfolio

 Work Authorization: Canadian Citizen | Eligible for U.S. TN Status (No H-1B required)

EDUCATION

University of British Columbia

September 2021 - April 2026

Bachelor of Applied Science - Integrated Engineering (Mechatronics Specialization), Co-op Program

cGPA: 4.0 (92%)

- Relevant Courses: Mechanics III: 95%, Automatic Control Systems: 88%, Electromechanics: 94%, Thermodynamics: 100%, Heat Transfer: 94%, Fluid Mechanics: 91%

SKILLS

- Mechanical Design: SolidWorks (CSWP Certified, FEA, Flow), CATIA/3DX, Fusion 360, AutoCAD, Ansys FEA/Fluent, GD&T
- Programming & Electronics: Python, C++, PLC Control (UR/KUKA), PCB design (Altium), ROS2, Git, MATLAB/Simulink
- Prototyping & Manufacturing: FDM/SLA 3D printing, CNC machining, laser/waterjet cutting

WORK EXPERIENCE

Hein Labs

December 2024 – January 2026

Engineering Project Lead - Mechatronics | SolidWorks, OpenCV, GD&T, ROS2, C++, Python, Altium

- Directed a **6-person engineering team** developing a robotic drug-analysis platform for BC's overdose crisis response, automating sample preparation and downstream analysis of **1500+ samples with 98% overall reliability**
- Designed, fabricated, tested, and iterated robotic lab-automation hardware in **SolidWorks/Fusion 360**, including a BLDC centrifuge, syringe-pump liquid handler, decapper, vial gripper, and robot transfer fixtures, reducing module costs from **\$1,500-\$5,000 to <\$100**.
- Produced **11 detailed CAD assemblies** and manufacturing drawings with **GD&T** for 3D-printed, CNC-machined, and waterjet-cut components
- Developed a multi-tiered control architecture using **PLC/Python** to orchestrate UR robotic arm motion and high-level workflows, integrated with **C++/ROS2** for hardware module automation and OpenCV error detection.
- Designed and fabricated **2 custom PCBs** using **Altium** for synchronized stepper and BLDC motor control

Hein Labs

April 2024 – December 2024

Mechatronics Engineering Intern | SolidWorks, RoboDK, Python, C++, Git, PLC, I/O Wiring

- Commissioned **4 industrial robotic arms** across **3 deployed self-driving chemistry lab platforms**, handling robot mounting, I/O wiring, networking, and end-to-end system integration while reducing scientists' hands-on time by **~80%**
- Transitioned hard-to-maintain **PLC-based robot sequences** to a reusable **Python control library** for **UR and KUKA** arms, making workflows easier to program, modify, and deploy
- Modeled **2 robotic platforms** in **SolidWorks** and simulated workflows in **RoboDK** to verify reachability, collision clearance, and lab module access before deployment, reducing workflow errors by **5x**

UBC AeroDesign

April 2022 - April 2026

Airfoils Team Lead | CATIA, SolidWorks, ANSYS, AutoCAD, GD&T, Manufacturing, Composites

- Led an **11-person subteam** designing and manufacturing a **120-inch heavy-lift RC aircraft** for SAE AeroDesign competition, overseeing wing design, large-scale assembly integration, manufacturing, and testing. Earned **2nd place overall in 2025**
- Designed main wing, ailerons, servo mounts, and linkage interfaces in **SolidWorks and CATIA V5** using DFMA and GD&T principles, allowing easy integration into overall assembly
- Optimized aircraft geometry through **SolidWorks FEA/Flow, ANSYS, and wind-tunnel validation**, improving lift performance by **~20%** and verifying structural stiffness before manufacturing
- Manufactured flight hardware using **3D printing, CNC machining, waterjet/laser cutting, and carbon-fiber prepreg layups**, standardizing assembly processes and reducing build errors by **70%**

PROJECTS

Robotic Tool Vending Machine (Haven)

September 2025 - April 2026

Project Lead | SolidWorks, Raspberry Pi, Klipper, KiCAD, Manufacturing & Prototyping

- Designed and validated a **Cartesian XY gantry** and rotary storage system in **SolidWorks** using **FEA** before fabrication.
- Fabricated and integrated prototypes using **3D printing, waterjet cutting, hand tools, and CNC machining**
- Developed **Raspberry Pi/Klipper motion control** with magnetic encoder feedback and **custom I2C multiplexer PCB** to synchronize gantry travel and storage indexing

Semi-Autonomous Aquatic Trash-Removal Robot

September 2024 - April 2025

Mechanical Lead | SolidWorks, 3D-printing, Waterjet, Machine Shop Tools

- Designed **aquatic mechanical assemblies** in **SolidWorks**, engineering custom dual-pontoon flotation chassis, structural supports, and adjustable propulsion mounts, winning **1st place** at the National CSME Design Competition
- Fabricated and integrated **sealed plywood, aluminum, and 3D-printed components** to assemble a watertight product optimized for saltwater operations